



ENVIRONMENT

SOCIAL ROBOT CO-DESIGN CANVASES

What is the robot's context of operation?
You can use the "Ecosystem" canvas to dive deeper into this topic.

Mercer General Hospital ED triage desk; integrates into the existing workflow without adding documentation time.

Where

What place?
Does it change?

Fixed triage workstations in the Emergency Department. Busy, noisy, fast-paced, with frequent interruptions, long queues and competing clinical priorities.

User(s)

Who is using the robot?

Emergency Department triage nurses; Nurse-in-Charge (Sister Patrice Alleyne).

When

What time of day?
Does it change?

Available 24 hours a day, every shift. Greatest benefit overnight, during seasonal surges, and other periods of high patient demand.

Secondary user(s)

Are there secondary users?
E.g. teachers that help students use a robot.

Emergency physicians; Observation Unit clinicians; hospital administrators; clinical IT staff; patients (indirect beneficiaries).

Data collection

Does the robot collect data from its environment?
How is it stored?

Collects demographics, symptoms, vital signs, medical history, AI predictions, confidence scores, overrides, timestamps and final decisions; stored securely with encryption and role-based access.



TRADE-OFF:

More data collection requires more attention to data security.

Simultaneous users

How many users should be able to use the robot simultaneously?



TRADE-OFF:

More simultaneous users requires a more sophisticated robot.



One primary user per workstation; each dashboard supports one triage nurse assessing one patient at a time, synchronised across the ED.

External sensors and actuators

Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

Sensors: blood pressure monitor, pulse oximeter, digital thermometer, heart rate monitor. Outputs: colour-coded urgency indicators, high-priority alerts, queue updates.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

Securely connects to Mercer EHR, patient registration, vital-sign monitors, hospital authentication and the clinical audit database; retrieves patient info, receives live vitals, records recommendations/overrides, and supports QI reporting.



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THRI paper:

